

PROJECT REPORT

UNSUPERVISED AGRO-ZONING BASED CROP RECOMMENDATION USING SOIL, YIELD, AND MARKET DATA

TEAM DETAILS

Team Name: AgroZone Analytics

Team Members

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1. PROBLEM BACKGROUND & MOTIVATION

Agricultural crop planning is a complex decision-making process influenced by soil fertility, climatic conditions, historical crop performance, and market demand. Traditional crop recommendation approaches commonly rely on supervised learning models that directly predict a single crop based on input features such as soil nutrients or rainfall. While straightforward, these models often ignore similarities across regions, fail to provide ranked alternatives, and rarely incorporate economic considerations such as market prices.

In real-world agricultural settings, reliable labeled data mapping specific soil samples to optimal crops is rarely available. Moreover, farmers and agricultural planners require multiple suitable options rather than a single prediction to support informed decision-making.

This project addresses these limitations by proposing a **context-aware, unsupervised agro-zoning-based crop recommendation system designed as a decision-support tool**. Instead of predicting crops directly, the system first discovers agro-zones using soil and climate similarity and then recommends crops based on historical yield performance and current market prices.

2. DATASET DESCRIPTION

This project uses three publicly available datasets sourced from Kaggle, each serving a distinct purpose in the system.

2.1 Soil and Climate Dataset (Agro-Zoning Input)

- **Dataset Name:** Crop Recommendation Dataset
- **Source:** Kaggle
- **Features Used:**
 - Nitrogen (N)
 - Phosphorus (P)
 - Potassium (K)
 - pH
 - Temperature
 - Humidity
 - Rainfall

The original crop label column was explicitly removed to ensure no supervised learning was performed. This dataset was used exclusively for unsupervised agro-zone discovery.

2.2 Historical Crop Production Dataset

- **Dataset Name:** Crop Production in India
- **Source:** Kaggle
- **Key Columns Used:**
 - State_Name
 - Season
 - Crop
 - Area
 - Production

A derived feature, **Yield**, was calculated as:

$$\text{Yield} = \text{Production} / \text{Area}$$

This dataset provides historical evidence of crop performance across different seasons.

2.3 Market Price Dataset

- **Dataset Name:** Agmarknet India Commodity Prices (Oct'24 – Aug'25)
- **Source:** Kaggle
- **Key Columns Used:**
 - Crop
 - Modal Price
 - Price Date

Prices were aggregated to compute average modal prices for each crop. These prices are used as a **relative economic indicator**, not as exact profit estimates.

3. METHODOLOGY & MODEL CHOICE

3.1 Agro-Zone Discovery (Unsupervised Learning)

To identify agro-zones, **K-Means clustering** was applied to standardized soil and climate features (N, P, K, pH, temperature, humidity, rainfall).

Preprocessing steps included:

- Removal of non-numeric and label columns
- Feature standardization using StandardScaler

The optimal number of clusters was determined using:

- Elbow Method
- Silhouette Score analysis

Based on both structural and interpretability considerations, **K = 5 agro-zones** was selected.

3.2 Agro-Zone Profiling

Each agro-zone was profiled by computing average soil and climate values. This step ensured interpretability by describing what each zone represents (e.g., high rainfall zone, mineral-rich soil zone).

	Zone_ID	N	P	K	temperature	humidity	ph	rainfall
0	0	25.743222	52.052632	23.151515	29.572003	65.557379	6.631671	74.132023
1	1	99.439068	41.508961	37.512545	25.635258	78.655925	6.437123	75.781013
2	2	49.041379	32.860345	34.091379	24.539337	85.108381	6.516712	168.067362
3	3	21.990000	133.375000	200.000000	23.240259	87.104305	5.977800	91.133304
4	4	28.697872	67.510638	45.561702	19.696806	23.326425	6.415449	98.500969

3.3 Recommendation Logic

For each season and crop, a recommendation score was computed using:

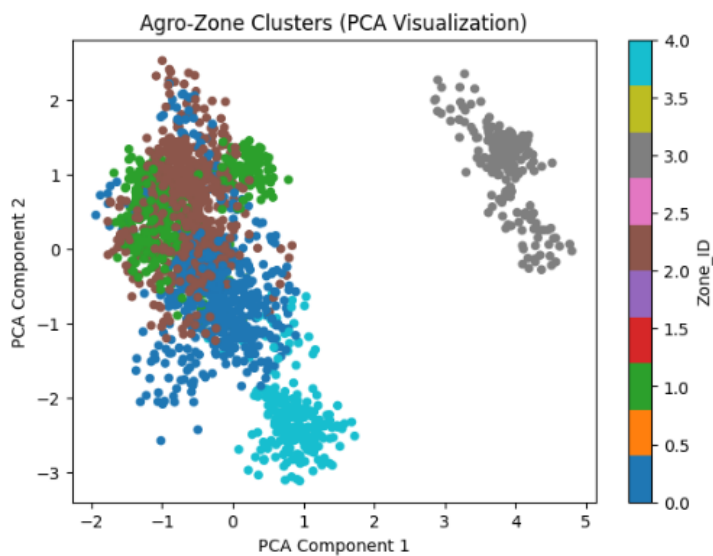
$$\text{Score} = 0.6 \times \text{Normalized_Yield} + 0.4 \times \text{Normalized_Market_Price}$$

- Yield was given higher importance as agronomic suitability is critical.
- Market price was included to incorporate economic relevance.
- The weights were fixed by design to maintain decision-support framing and avoid profit optimization.

4. EXPERIMENTS & RESULTS

4.1 Agro-Zone Identification Results

Five distinct agro-zones were discovered based on soil and climate similarity. Visualization using PCA confirmed reasonable separation between clusters.



4.2 Season-wise Crop Ranking Results

Crops were ranked separately for each season (Kharif, Rabi, Summer) based on the final score. For each season, the **Top-3 crops** were selected to keep recommendations practical and actionable.

Season	Crop	Yield	Avg_Modal_Price	Norm_Yield	Norm_Price	Score	Rank
Kharif	Garlic	2.400661	12773.246005	0.455363	1.000000	0.673218	1.0
Kharif	Mango	4.785274	4508.939220	0.933987	0.241011	0.656797	2.0
Kharif	Brinjal	5.114165	2317.671478	1.000000	0.039766	0.615906	3.0
Rabi	Garlic	2.965592	12773.246005	0.418269	1.000000	0.650962	1.0
Rabi	Banana	5.118355	2573.836778	1.000000	0.063292	0.625317	2.0
Rabi	Brinjal	5.114159	2317.671478	0.998866	0.039766	0.615226	3.0
Summer	Banana	5.118355	2573.836778	1.000000	0.063292	0.625317	1.0
Summer	Brinjal	5.118355	2317.671478	1.000000	0.039766	0.615906	2.0
Summer	Maize	2.101925	2210.436403	0.199427	0.029917	0.131623	3.0

4.3 Zone-wise Recommendation Output

Since agro-zones are not geographically mapped, the same season-wise Top-3 crops are applied uniformly across all agro-zones. This avoids false precision and maintains methodological honesty.

The final output includes:

- Agro-Zone ID
- Season
- Top-3 crops
- Yield, price, and score for transparency

N

50

P

40

K

40

Temp (°C)

25

Humidity (%)

70

pH

6.5

Rainfall (mm)

150

Season

Kharif

Get Recommendation...

Identified Agro-Zone: 2

Zone_ID	Season	Rank	Crop	Yield	Avg_Modal_Price	Score	
18	2	Kharif	1.0	Garlic	2.400661	12773.246005	0.673218
19	2	Kharif	2.0	Mango	4.785274	4508.939220	0.656797
20	2	Kharif	3.0	Brinjal	5.114165	2317.671478	0.615906

5. INSIGHTS & BUSINESS INTERPRETATION

Key insights derived from the system include:

- Agro-zones effectively capture shared soil and climate characteristics.
- Yield-based ranking ensures agronomic feasibility.
- Market price integration improves economic relevance.
- Providing multiple ranked crops supports better decision-making compared to single-crop prediction systems.

This system can assist farmers, agricultural planners, and policymakers in understanding suitable crop options for different soil-climate contexts and seasons.

6. LIMITATIONS & FUTURE SCOPE

6.1 Limitations

- Soil data is not plot-level or region-specific.
- Market prices may vary significantly over time.
- Recommendations are advisory and should not be treated as guarantees.

6.2 Future Scope

- Incorporating region-specific price trends
 - Adding risk and price volatility analysis
 - Including crop rotation constraints
 - Integrating real-time weather data
 - Deploying as a lightweight web or mobile interface
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7. CONCLUSION

This project demonstrates how unsupervised learning and recommendation systems can be combined to create a realistic, interpretable, and economically aware crop recommendation framework. By focusing on agro-zoning rather than direct prediction, the system aligns more closely with real-world agricultural decision-making and provides meaningful, ranked recommendations.